

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12384605
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Xu; Bin

Lock catch structure and storage box of chess pieces

Abstract

A lock catch structure and a storage box of chess pieces are provided. The lock catch structure includes a connecting buckle, a first fixing component, a lock cylinder, and a second fixing component. The connecting buckle is rotatably connected to the first fixing component, and the first fixing component is connected to a first end of a locked object. The lock cylinder is inserted into the second fixing component, and the second fixing component is connected to a second end of the locked object. The connecting buckle defines an accommodating cavity for accommodating the first fixing component and the second fixing component, the first fixing component and the second fixing component lock the first end of the locked object and the second end of the locked object.

Inventors:	Xu; Bin (Shenzhen, CN)
Applicant:	A&A INTELLIGENT (SHENZHEN) CO., LTD. (Shenzhen, CN)
Family ID:	89348702
Assignee:	A&A INTELLIGENT (SHENZHEN) CO., LTD. (Shenzhen, CN)
Appl. No.:	18/434740
Filed:	February 06, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240425244 A1	Dec. 26, 2024

Foreign Application Priority Data

CN	202321593475.7	Jun. 20, 2023
----	----------------	---------------

Publication Classification

Int. Cl.: B65D43/22 (20060101); B65D51/00 (20060101)

U.S. Cl.:

CPC B65D43/22 (20130101); B65D51/00 (20130101); B65D2251/1058 (20130101)

Field of Classification Search

CPC: E05C (17/50); E05B (7/0005); B65D (2251/1058); B65D (51/00); B65D (43/22)

USPC: 206/315.14; 292/246

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1184439	12/1915	Giles	292/DIG.20	E05C 17/50
2829912	12/1957	Koch	70/73	E05C 19/14
3030137	12/1961	Tomkinson	292/113	E05C 19/14
4540206	12/1984	Frame	292/DIG.60	E05C 19/14
4705308	12/1986	Bisbing	292/108	E05C 19/14
4801165	12/1988	Pyle	292/249	E05C 19/14
5271649	12/1992	Gromotka	292/DIG.40	E05C 19/14
5478125	12/1994	Gromotka	292/113	E05C 19/14
6257631	12/2000	Sokurenko	292/DIG.49	E05C 19/14
6457750	12/2001	Sokurenko	292/113	E05C 19/14
7017955	12/2005	Chiang	292/113	E05C 19/14
D548565	12/2006	Vickers	D8/331	N/A
7568739	12/2008	Lee	292/85	B25H 3/02
7654589	12/2009	Vickers	292/DIG.31	E05B 13/002
9663969	12/2016	Lila	N/A	E05B 17/0058
10309134	12/2018	Prehn	N/A	E05C 19/14
10392830	12/2018	Helsley	N/A	E05C 19/14
D889932	12/2019	Zeng	D8/331	N/A
D900588	12/2019	Rolfs	D8/331	N/A
D993743	12/2022	Zhou	D8/341	N/A
2012/0139296	12/2011	Wilkins	296/193.11	E05C 19/14
2014/0265367	12/2013	Lila	292/346	E05C 19/12

Primary Examiner: Grano; Ernesto A

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a technical field of mechanical locks, and in particular to a lock catch structure and a storage box of chess pieces.

BACKGROUND

(2) In the prior art, various types of mechanical locks are provided for locking doors, cases,

cabinets, boxes, etc. Generally, structures of the various types of the mechanical locks are complex and require corresponding keys for unlocking, so that if there is an occasion just requiring a temporary lock, a temporary lock structure requiring a corresponding key for unlocking to avoid a locked object from being accidentally opened brings much inconvenience.

(3) A storage box of chess pieces requires the temporary lock, after users finish chess, the chess pieces need to be stored in the storage box, in order to prevent the chess pieces stored in the storage box from losing, the storage box needs to be temporarily locked through a lock structure, so that the storage box is prevented from being accidentally opened. However, when a current storage box of chess pieces is locked, since there is a relatively large gap between a lock structure of the current storage box when the lock structure works to lock the current storage box, the lock structure is not stable, and after using for a long time, a locking function may be lost due to abrasion; if there is a relatively small gap between the lock structure of the current storage box when the lock structure works to lock the current storage box, after using for a long time, dust easily enters the lock structure to form dirt, so that the lock structure has much difficulty in working to lock the current storage box. Therefore, how to enable a lock catch structure to have stability for using for a long time and avoid the lock catch structure from entering the dust to cause locking difficulty is a problem to be solved.

SUMMARY

(4) A first purpose of the present disclosure is to provide a lock catch structure for solving a technical problem about how to enable the lock catch structure to have stability for using for a long time and avoid the lock catch structure from entering the dust to cause locking difficulty.

(5) In order to solve above technical problems, the lock catch structure is provided, including an upper catch assembly and a lower match assembly. The upper catch assembly includes a connecting buckle and a first fixing component, the connecting buckle is rotatably connected to the first fixing component, and the first fixing component is connected to a first end of a locked object. The lower match assembly includes a lock cylinder and a second fixing component, the lock cylinder is inserted into the second fixing component, and the second fixing component is connected to a second end of the locked object. The first fixing component includes a rotating portion and a connecting portion, the rotating portion is connected to the connecting buckle, and the connecting portion is disposed away from the rotating portion. The connecting buckle defines an accommodating cavity, the connecting buckle rotates with respect to the rotating portion, so as to accommodate the first fixing component and the second fixing component in the accommodating cavity, and the lock cylinder is inserted into the connecting portion, so that the first fixing component and the second fixing component lock the first end of the locked object and the second end of the locked object.

(6) Furthermore, the connecting portion defines a lock hole, when the connecting portion is located in the accommodating cavity, the lock cylinder extends from the second fixing component into the lock hole.

(7) Furthermore, the connecting buckle defines a positioning hole, the lock cylinder includes an inserting portion and a positioning portion, the positioning portion is disposed away from the inserting portion, the inserting portion is inserted into the lock hole, the positioning portion is inserted into the positioning hole, so as to lock the second fixing component to the connecting buckle with respect to the first fixing component.

(8) Furthermore, the positioning portion includes a ball head, when the connecting buckle rotates with respect to the rotating portion, the ball head slides into the positioning hole of the connecting buckle.

(9) Furthermore, the upper catch assembly further includes a cover plate, a first side surface and a second side surface are formed on two sides of the connecting buckle, the cover plate is disposed on the first side surface, and the accommodating cavity is defined between the cover plate and the second side surface, when the first fixing component or the second fixing component is

accommodated in the accommodating cavity, one side of the first fixing component or one side of the second fixing component is in contact with the second side surface.

(10) Furthermore, the cover plate is detachably connected to the connecting buckle, and the cover plate is capable of being detached from the first side surface.

(11) Furthermore, the upper catch assembly further includes a rotating shaft, the rotating shaft passes through the rotating portion and the connecting buckle, the first fixing component includes a first bottom plate and a first protrusion protruding from the first bottom plate, the lock hole is defined on the first protrusion, when the connecting buckle rotates with respect to the rotating shaft to accommodate the first fixing component in the accommodating cavity, one side, close to the first protrusion, of the first bottom plate is in contact with the second side surface.

(12) Furthermore, the second fixing component includes a second bottom plate and a second protrusion protruding from the second bottom plate, the lock cylinder passes through the second protrusion, when the connecting buckle rotates with respect to the rotating shaft to accommodate the second fixing component in the accommodating cavity, one side, close to the second protrusion, of the second bottom plate is in contact with the second side surface, and the lock cylinder is directly inserted into the lock hole.

(13) A second purpose of the present disclosure is to provide a storage box of chess pieces for solving a technical problem about how to enable the storage box of the chess pieces to have stability for using for a long time and avoid a lock catch structure of the storage box of the chess pieces from entering the dust to cause locking difficulty.

(14) In order to solve above technical problems, the storage box of the chess pieces is provided, including the lock catch structure as foregoing, a first frame plate, and a second frame plate. The upper catch assembly is detachably connected to the first frame plate, and the lower match assembly is detachably connected to the second frame plate.

(15) Furthermore, the first frame plate and the second plate are overlapped to form a storage cavity for storing the chess pieces, where the first fixing component and the second fixing component lock the first frame plate and the second frame plate.

(16) Embodiments of the present disclosure have following beneficial effects.

(17) In a first embodiment, the lock catch structure is provided, when the connecting buckle rotates with respect to the rotating portion, the first fixing component and the second fixing component are accommodated in the accommodating cavity, and the lock cylinder is inserted into the connecting portion, so that the first fixing component and the second fixing component lock the locked object. In this way, stability enabling the lock catch structure to be used for a long time is improved and a problem in the prior art that stability of a current lock catch may be reduced after a long-term use is solved.

(18) In a second embodiment, the storage box of the chess pieces is provided, the storage box of the chess pieces includes the lock catch structure provided in the first embodiment, the first frame plate, and the second frame plate. The upper catch assembly is detachably connected to the first frame plate, and the lower match assembly is detachably connected to the second frame plate. In this way, the first fixing component and the second fixing component lock the first frame plate and the second frame plate, thereby preventing the first frame plate and the second frame plate from being accidentally opened and further preventing the chess pieces stored in the storage cavity from losing.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) In order to more clearly illustrate embodiments of the present disclosure or technical solutions in the prior art, drawings that need to be used in the embodiments or the prior art are briefly

described below, and it is obvious that accompanying drawings in following description are merely some embodiments of the present disclosure, and those who skilled in the art may obtain other drawings according to these drawings without involving any inventive effort.

(2) FIG. 1 is a schematic diagram of a lock catch structure presented in a first angle according to a first embodiment of the present disclosure.

(3) FIG. 2 is a schematic diagram of the lock catch structure presented in a second angle according to the first embodiment of the present disclosure.

(4) FIG. 3 is an exploded schematic diagram of the lock catch structure according to the first embodiment of the present disclosure.

(5) FIG. 4 is a schematic diagram of a connecting buckle presented in a first angle according to the first embodiment of the present disclosure.

(6) FIG. 5 is a schematic diagram of the connecting buckle presented in a second angle according to the first embodiment of the present disclosure.

(7) FIG. 6 is a structural schematic diagram of a first fixing component according the first embodiment of the present disclosure.

(8) FIG. 7 is a structural schematic diagram of a second fixing component according to the first embodiment of the present disclosure.

(9) FIG. 8 is a structural schematic diagram of a storage box of chess pieces according a second embodiment of the present disclosure.

(10) Reference numerals in the drawings: **100**, lock catch structure; **110**, upper catch assembly; **111**, connecting buckle; **1111**, accommodating cavity; **1112**, positioning hole; **1113**, first side surface; **1114**, second side surface; **112**, first fixing component; **1121**, rotating portion; **1122**, connecting portion; **11221**, lock hole; **1123**, first bottom plate; **1124**, first protrusion; **113**, cover plate; **114**, rotating shaft; **120**, lower match assembly; **121**, lock cylinder; **1211**, inserting portion; **1212**, positioning portion; **122**, second fixing component; **1221**, second bottom plate; **1222**, second protrusion; **200**, storage box of chess pieces; **210**, first frame plate; **220**, second frame plate; **230**, storage cavity.

DETAILED DESCRIPTION

(11) In order to facilitate understanding of the present disclosure, the present disclosure is described more fully hereinafter with reference to accompanying drawings. Preferred embodiments of the present disclosure are given in the accompanying drawings. However, the present disclosure may be implemented in many different forms and is not limited to embodiments described herein. On the contrary, a purpose of providing these embodiments is to make the understanding of a content of the present disclosure more thorough and comprehensive.

(12) It should be noted that when an element is referred to as being “fixed to” another element, it may be directly on another element or intermediate elements may also be provided. When an element is considered to be “connected to” another element, it may be directly connected to another element or intermediate elements may also be provided. Terms “vertical”, “horizontal”, “left”, “right”, etc. used herein are for illustrative purposes only.

(13) Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by those who skilled in the art of the present disclosure. A terminology used herein in the specification of the present disclosure is for a purpose of describing particular embodiments only and is not intended to limit the present disclosure. As used herein, a term “and/or” includes any and all combinations of one or more of associated listed items.

First Embodiment

(14) Please refer to FIGS. 1-7, the first embodiment of the present disclosure provides a lock catch structure **100**, including an upper catch assembly **110** and a lower match assembly **120**. The upper catch assembly **110** includes a connecting buckle **111** and a first fixing component **112**, the connecting buckle **111** is rotatably connected to the first fixing component **112**, and the first fixing component **112** is connected to a first end of a locked object. The lower match assembly **120**

includes a lock cylinder **121** and a second fixing component **122**, the lock cylinder **121** is inserted into the second fixing component **122**, and the second fixing component **122** is connected to a second end of the locked object. The first fixing component **112** includes a rotating portion **1121** and a connecting portion **1122**, the rotating portion **1121** is connected to the connecting buckle **111**, and the connecting portion **1122** is disposed away from the rotating portion **1121**. The connecting buckle **111** defines an accommodating cavity **1111**, the connecting buckle **111** rotates with respect to the rotating portion **1121**, so as to accommodate the first fixing component **112** and the second fixing component **122** in the accommodating cavity **1111**, and the lock cylinder **121** is inserted into the connecting portion **122**, so that the first fixing component **112** and the second fixing component **112** lock the first end of the locked object and the second end of the locked object. In the specific application, when the connecting buckle **111** rotates with respect to the rotating portion **1121**, the connecting portion **1122** of the first fixing component **112** is gradually accommodated in the accommodating cavity **1111**, and in this process, since the connecting portion **1122** is disposed away from the rotating portion **1121**, when the connecting buckle **111** rotates with respect to the rotating portion **1121**, a part, close to the rotating portion **1121**, of the first fixing component **112** is first accommodated in the accommodating cavity **1111**, and then the accommodating cavity **1111** gradually accommodates the connecting portion **1122**, when the connecting portion **1122** starts to be gradually accommodated into the accommodating cavity **1111**, the second fixing component **122** abuts against the connecting portion **1122** to accommodate into the accommodating cavity **230** along with the connecting portion **1122**, when the first fixing component **112** and the second fixing component **122** are completely accommodated into the accommodating cavity **1111**, the lock cylinder **121** passing through the second fixing component **122** is inserted into the connecting portion **1122**, since the first fixing component **112** is connected to the first end of the locked object, the second fixing component **122** is connected to the second end of the locked object, the first fixing component **112** and the second fixing component **122** lock the first end of the locked component and the second of the locked component to complete temporary locking. In this way, the lock catch structure **100** quickly and reliably locks the locked object temporarily.

(15) In one implementation, the connecting portion **1122** defines a lock hole **11221**, when the connecting portion **1122** is located in the accommodating cavity **1111**, the lock cylinder **121** extends from the second fixing component **122** into the lock hole **11221**. In specific application, in order to improve stability of the lock catch structure **100** after working to lock, the lock hole **11221** is defined on the connecting portion **1122**, when the connecting portion **1122** starts to enter the accommodating cavity **1111**, the lock cylinder **121** passing through the second fixing component **122** starts to engage with the lock hole **11221**, so that when the connecting portion **1122** of the first fixing component **112** is completely accommodated into the accommodating cavity **1111**, the first fixing component **112** is fastened with the second fixing component **122**, so that the locked object is avoided from being accidentally opened.

(16) In one implementation, the connecting buckle **111** defines a positioning hole **1112**, the lock cylinder **121** includes an inserting portion **1211** and a positioning portion **1212**, the positioning portion **1212** is disposed away from the inserting portion **1211**, the inserting portion **1211** is inserted into the lock hole **11221**, the positioning portion **1212** is inserted into the positioning hole **1112**, so as to lock the second fixing component **122** to the connecting buckle **111** with respect to the first fixing component **112**. In specific application, since the lock cylinder **121** includes the inserting portion **1211** and the positioning portion **1212**, the inserting portion **1211** is configured to insert into the lock hole **11221**, and the positioning portion **1212** is configured to insert into the positioning hole **1112**, and when the second fixing component **122** enters the accommodating cavity **1111**, the positioning hole **1112** is formed on the connecting buckle **111** corresponding to the positioning portion **1212**, so that the positioning portion **1212** is inserted into the positioning hole **1112**. In this way, the first fixing component **112** is fastened to the connecting buckle **111**, so that the second fixing component **122** is locked on the connecting buckle **111** with respect to the first

fixing component **112**.

(17) In one implementation, the positioning portion **1212** includes a ball head (not shown in the drawings), when the connecting buckle **111** rotates with respect to the rotating portion **1121**, the ball head slides into the positioning hole **1112** of the connecting buckle **111**. In specific application, in order to easily abut the first fixing component **112** against the connecting buckle **111**, when the connecting buckle **111** is in contact with the positioning portion **1212**, the ball head of the positioning portion **1212** slides into the positioning hole **1112** of the connecting buckle **111**. In this way, through using the ball head to be in contact with the connecting buckle **111**, a contact area between the positioning portion **1212** and the connecting buckle **111** is reduced, so that the first fixing component **112** easily abuts against the connecting buckle **111**.

(18) In one implementation, the upper catch assembly **110** further includes a cover plate **113**, a first side surface **1113**, and a second side surface **1114** are formed on two sides of the connecting buckle **111**, the cover plate **113** is disposed on the first side surface **1113**, and the accommodating cavity **1111** is defined between the cover plate **113** and the second side surface **1114**, when the first fixing component **112** or the second fixing component **122** is accommodated in the accommodating cavity **1111**, one side of the first fixing component **112** or one side of the second fixing component **122** is in contact with the second side surface **1114**. In specific application, since the cover plate **113** covers the first side surface **1113**, the accommodating cavity **1111** is formed between the cover plate **113** and the second side surface **1114**, and a fastening position of the first fixing component **112** and the second fixing component **122** are located in the accommodating cavity **1111**, the cover plate **113** blocks dust in air, so that the dust in the air cannot enter the accommodating cavity **1111**. In this way, a cooperation gap between the first fixing component **112** and the second fixing component **122** are prevented from accumulating dust to form dirt to affect fluency of fastening. In addition, when the first fixing component **112** or the second fixing component **122** is accommodated in the accommodating cavity **1111**, the one side of the first fixing component **112** or the one side of the second fixing component **122** is in contact with the second side surface **1114**, the dust is also well prevented from entering the accommodating cavity **1111**.

(19) In one implementation, the cover plate **113** is detachably connected to the connecting buckle **111**, and the cover plate **113** is capable of being detached from the first side surface **1113**. In specific application, the cover plate **113** is detachably connected to the connecting buckle **111**, so that the cover plate **113** is capable of being replaced, different patterns are printed or engraved on one side, facing away from the accommodating cavity **1111**, of the cover plate **113** to improve ornamental value or attractiveness of the lock catch structure **100**, and in addition, the cover plate **113** is detached from the first side surface **1113** through fasteners, so that the fasteners are prevented from being exposed on the one side, facing away from the accommodating cavity **1111**, of the cover plate **113** to affect the attractiveness of the lock catch structure **100**.

(20) In one implementation, the upper catch assembly **110** further includes a rotating shaft **114**, the rotating shaft **114** passes through the rotating portion **1121** and the connecting buckle **111**, the first fixing component **112** includes a first bottom plate **1123** and a first protrusion **1124** protruding from the first bottom plate **1123**, the lock hole **11221** is defined on the first protrusion **1124**, when the connecting buckle **111** rotates with respect to the rotating shaft **114** to accommodate the first fixing component **112** in the accommodating cavity, one side, close to the first protrusion **1124**, of the first bottom plate **1123** is in contact with the second side surface **1114**. In specific application, when the connecting buckle **111** rotates with respect the rotating shaft **114** to enable the first fixing component **112** to be accommodated in the accommodating cavity **1111**, the one side, close to the first protrusion **1124**, of the first bottom plate **1123** is in contact with the second side surface **1114**, so that the lock hole **11221** of the first protrusion **1124** arrives a predetermined position through contact and cooperation between the one side, close to the first protrusion **1124**, of the first bottom plate **1123** and the second side surface **1114**.

(21) In one implementation, the second fixing component **112** includes a second bottom plate **1221**

and a second protrusion **1222** protruding from the second bottom plate **1221**, the lock cylinder **121** passes through the second protrusion **1222**, when the connecting buckle **111** rotates with respect to the rotating shaft **114** to accommodate the second fixing component **122** in the accommodating cavity **1111**, one side, close to the second protrusion **1222**, of the second bottom plate **1221** is in contact with the second side surface **1114**, and the lock cylinder **121** is directly inserted into the lock hole **11221**. In specific application, when the connecting buckle **111** rotates with respect to the rotating shaft **114** to enable the second fixing component **122** to be accommodated in the accommodating cavity **1111**, the one side, close to the second protrusion **1222**, of the second bottom plate **1221** is in contact with the second side surface **1114**, so that the lock cylinder **121** passing through the second protrusion **1222** arrives the same position as the lock hole **11221** through contact and cooperation between the one side, close to the second protrusion **1222**, of the second bottom plate **1221** and the second side surface **1114**, so that the lock cylinder **121** is directly inserted into the lock hole **11221**, thereby improving docking precision of locking.

Second Embodiment

(22) The second embodiment is different from the subject matter set forth in the first embodiment. Details can be referred to following.

(23) Please refer to FIGS. **1**, **2**, and **8**, the second embodiment of the present disclosure provides a storage box **200** of chess pieces, including the lock catch structure **100**, a first frame plate **210**, and a second frame plate **220**. The upper catch assembly **110** is detachably connected to the first frame plate **210**, and the lower match assembly **120** is detachably connected to the second frame plate **220**. In specific application, the first fixing component **112** of the upper catch assembly **110** is detachably connected to the first frame plate **210**, and the second fixing component **122** of the lower match assembly **120** is detachably connected to the second frame plate **220**, thereby improving convenience of assembling the storage box **200** for the chess pieces with the lock catch structure **100**.

(24) In one implementation, the first frame plate **210** and the second plate **220** are overlapped to form a storage cavity **230** for storing the chess pieces, where the first fixing component **112** and the second fixing component **122** lock the first frame plate **210** and the second frame plate **220**. In specific application, the first frame plate **210** and the second plate **220** are overlapped to form the storage cavity **230** for storing the chess pieces, where the first fixing component **112** and the second fixing component **122** lock the first frame plate **210** and the second frame plate **220**, since the lock catch structure **100** is capable of temporarily locking the storage box **200** for the chess pieces through the first fixing component **112** and the second fixing component **122**, the first frame plate **210** and the second frame plate **220** of the storage box **200** for the chess pieces are avoided from being accidentally separated, causing the chess pieces therein to accidentally drop from the storage cavity **230**. When the chess pieces in the storage box **200** of the chess pieces need to be taken out, only the connecting buckle **111** needs to be pried to rotate with respect the first fixing component **112**, so that the first fixing component **112** and the second fixing component **122** are separated, and the first frame plate **210** and the second frame plate **220** are opened to take out the chess pieces in the storage box **200** of the chess pieces. Therefore, using the lock catch structure **100** provided by the present disclosure in the storage box **200** of the chess pieces may improve stability when locking the lock catch structure **100** and convenience when unlocking the lock catch structure **100**.

(25) Above embodiments only express several embodiments of the present disclosure, and description thereof is more specific and detailed, but cannot be understood as a limitation on a protection scope of the present disclosure. It should be noted that, for those who skilled in the art, several variations and improvements may be made without departing from a concept of the present disclosure, and these are all within the protection scope of the present disclosure. Therefore, the protection scope of the present disclosure shall be subject to the appended claims.

Claims

1. A lock catch structure, comprising: an upper catch assembly; and a lower match assembly; wherein the upper catch assembly comprises a connecting buckle and a first fixing component, the connecting buckle is rotatably connected to the first fixing component, and the first fixing component is connected to a first end of a locked object; the lower match assembly comprises a lock cylinder and a second fixing component, the lock cylinder is inserted into the second fixing component, and the second fixing component is connected to a second end of the locked object; the first fixing component comprises a rotating portion and a connecting portion, the rotating portion is connected to the connecting buckle, and the connecting portion is disposed away from the rotating portion; the connecting buckle defines an accommodating cavity, the connecting buckle rotates with respect to the rotating portion, so as to accommodate the first fixing component and the second fixing component in the accommodating cavity, and the lock cylinder is inserted into the connecting portion, so that the first fixing component and the second fixing component lock the first end of the locked object and the second end of the locked object.
2. The lock catch structure according to claim 1, wherein the connecting portion defines a lock hole, when the connecting portion is located in the accommodating cavity, the lock cylinder extends from the second fixing component into the lock hole.
3. The lock catch structure according to claim 2, wherein the connecting buckle defines a positioning hole, the lock cylinder comprises an inserting portion and a positioning portion, the positioning portion is disposed away from the inserting portion, the inserting portion is inserted into the lock hole, the positioning portion is inserted into the positioning hole, so as to lock the second fixing component to the connecting buckle with respect to the first fixing component.
4. The lock catch structure according to claim 3, wherein the positioning portion comprises a ball head, when the connecting buckle rotates with respect to the rotating portion, the ball head slides into the positioning hole of the connecting buckle.
5. The lock catch structure according to claim 2, wherein the upper catch assembly further comprises a cover plate, a first side surface and a second side surface are formed on two sides of the connecting buckle, the cover plate is disposed on the first side surface, and the accommodating cavity is defined between the cover plate and the second side surface, when the first fixing component or the second fixing component is accommodated in the accommodating cavity, one side of the first fixing component or one side of the second fixing component is in contact with the second side surface.
6. The lock catch structure according to claim 5, wherein the cover plate is detachably connected to the connecting buckle, and the cover plate is capable of being detached from the first side surface.
7. The lock catch structure according to claim 5, wherein the upper catch assembly further comprises a rotating shaft, the rotating shaft passes through the rotating portion and the connecting buckle, the first fixing component comprises a first bottom plate and a first protrusion protruding from the first bottom plate, the lock hole is defined on the first protrusion, when the connecting buckle rotates with respect to the rotating shaft to accommodate the first fixing component in the accommodating cavity, one side, close to the first protrusion, of the first bottom plate is in contact with the second side surface.
8. The lock catch structure according to claim 7, wherein the second fixing component comprises a second bottom plate and a second protrusion protruding from the second bottom plate, the lock cylinder passes through the second protrusion, when the connecting buckle rotates with respect to the rotating shaft to accommodate the second fixing component in the accommodating cavity, one side, close to the second protrusion, of the second bottom plate is in contact with the second side surface, and the lock cylinder is directly inserted into the lock hole.
9. A storage box of chess pieces, comprising: the lock catch structure according to claim 1; a first

frame plate; and a second frame plate; wherein the upper catch assembly is detachably connected to the first frame plate, and the lower match assembly is detachably connected to the second frame plate.

10. The storage box of the chess pieces according to claim 9, wherein the first frame plate and the second plate are overlapped to form a storage cavity for storing the chess pieces, where the first fixing component and the second fixing component lock the first frame plate and the second frame plate.
